

Supplementary File 3: Nuclei Area Quantification

for “Four-Fold Expansion Microscopy of Cultured Cells” by Corban Swain, Sapna Sinha, and Ed Boyden

Jupyter Notebook (converted to .pdf) containing ImageJ commands and Python script for quantification of nuclei area from images.

1. Setting Up Repository and Opening the Notebook

1. You will need to install and set up the following software before continuing:

- ☐ **git** <https://git-scm.com/>
- ☐ **mamba** <https://mamba.readthedocs.io>
- ☐ **Fiji (ImageJ)** <https://imagej.net/software/fiji/downloads> (version 1.54p was used)

2. Clone the **ExM-Demo-Figures** repository (<https://github.com/CorbanSwain/ExM-Demo-Figures>) into a directory (e.g. ~/ExM-Demo-Figures) on your computer. Then initialize and update the submodules.

```
cd ~
git clone https://github.com/CorbanSwain/ExM-Demo-Figures ~/ExM-Demo-Figures
git submodule update --init
```

3. Set up the Python environment according to the environment configuration provided in the repository.

```
cd ~/ExM-Demo-Figures/envs/
mamba create -f exm-demo-fig-py3.12-env.yml
```

4. Then run JupyterLab and open this notebook via the JupyterLab interface.

```
cd ~/ExM-Demo-Figures/source
mamba activate exm-demo-fig-py3.12-env
jupyter lab supplementary_file_03-nuc-area.ipynb
```

2. Segmentation and Measurement of Images in Fiji

Follow these instructions to process the images for nuclei area quantification.

1. Open Image

- Run Fiji and open the image located at ~/ExM-Demo-Figures/data/nuc-size-analysis/expanded-hex-cells-a.tif
This is fluorescent microscope image of expanded HEK cells taken with a 40x objective on a Nikon CSU-W1. The DAPI channel has already been separated as its own image file.
- This image has embedded scale information (i.e. 5.8954 px/micron). If an image being analyzed does not have scale information, it should first be set via the Analyze > Set Scale... dialog.
- Image > Adjust > Brightness/Contrast...
 - Click Reset
 - Click Auto
- LUT Toolbar Button > Grays

2. Duplicate Image

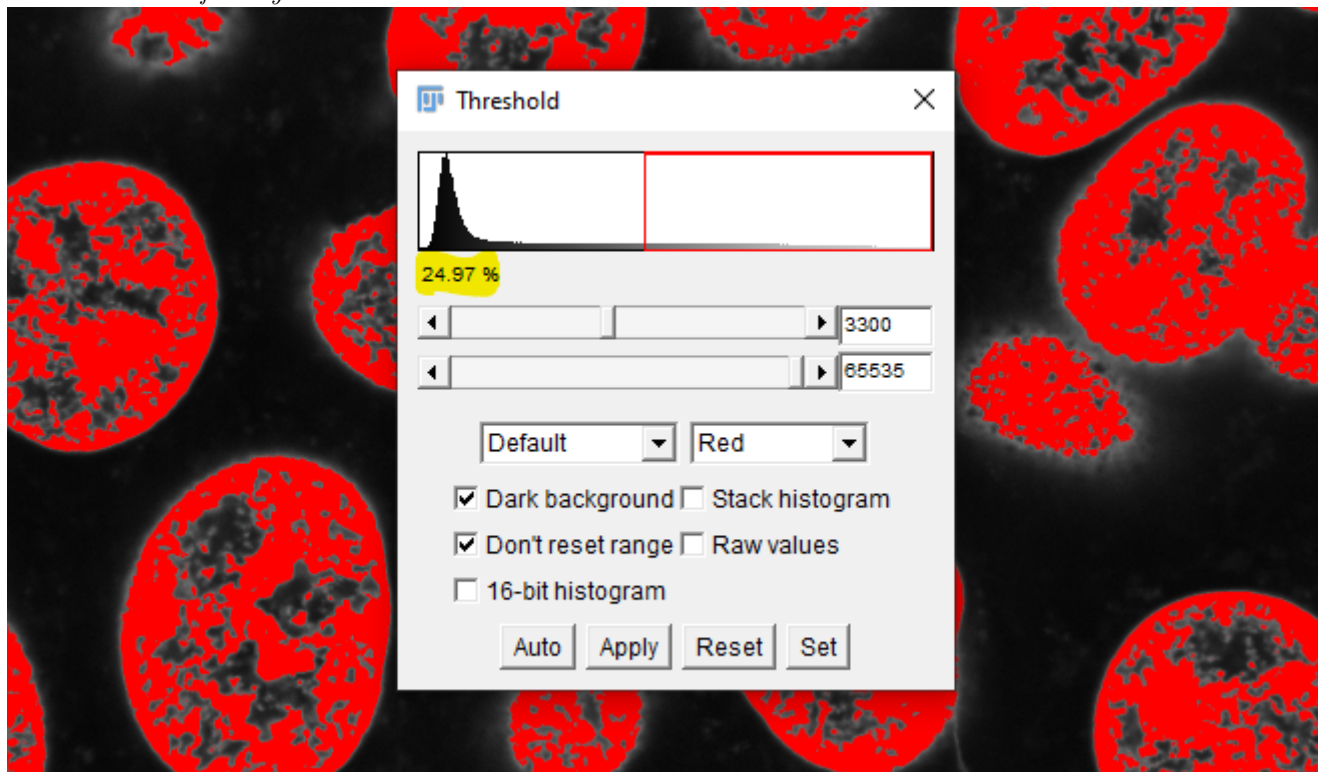
- Image > Duplicate...
- Title: = expanded-hek-cells-a-MASK.tif
- Perform the following steps on the expanded-hek-cells-a-MASK.tif file.

3. Histogram Normalization

- Process > Enhance Local Contrast (CLAHE)
- blocksize: = 255 for expanded samples (63 for unexpanded)
- histogram bins: = 256
- maximum slope: = 2.00
- mask: = *None*
- fast = NO

4. Thresholding

- Image > Adjust > Threshold...
- lower threshold value: Adjust until approximately 25% of pixels are selected. *This target percentage may need to be adjusted based on cell confluency.*



- upper threshold value: maximum
- method: Default
- color: Red

- Dark background = YES
 - Don't reset range = YES
 - 16-bit histogram = NO
 - Stack histogram = NO
 - Raw values = NO
 - Finally, click Apply.
5. **Binary Closing**
- Process > Binary > Options...
 - Iterations: = 4 for expanded samples (1 for unexpanded)
 - Count: = 1
 - Black background = YES
 - Pad edges when eroding = YES
 - EDM output: = 8-bit
 - Do: = Close
6. **Binary Fill Holes**
- Process > Binary > Fill Holes
7. **Watershed Segmentation**
- Process > Binary Watershed
8. **Save Mask**
- File > Save...
 - Save in: ~/ExM-Demo-Figures/data/nuc-size-analysis/
 - File name: expanded-hek-cells-a-MASK.tif
9. **Analyze**
- Analyze > Analyze Set Measurements...
 - Area = YES
 - *all others* = NO
 - Decimal Places: = 3
 - Analyze > Analyze Particles...
 - Size (micron²): = 500-4000 for expanded samples (31-250 for unexpanded)
 - Pixel units = NO
 - Circularity: = 0.5-1.00
 - Show: = Overlay
 - Display results = YES
 - Clear results = YES
 - Summarize = NO
 - Add to Manager = NO
 - Record starts = NO
 - Exclude on edges = YES
 - Include holes = YES
 - Overlay = YES
 - Composite ROIs = NO
10. **Save Results**

- Select the window **Results**
- **File > Save As** (Within the Results window, not the Fiji toolbar)
- Save in: ~/ExM-Demo-Figures/data/nuc-size-analysis/
- File name: **expanded-hek-cells-a-RESULTS.csv**

11. Transfer Regions

- Select the window **expanded-hek-cells-a-MASK.tif**
- **Image > Overlay > To ROI Manager** *the ROI Manager window should now be open.*
- Select the window **expanded-hek-cells-a.tif**
- Select the window **ROI Manager**
- Click the first region (i.e. 1-1) in the ROI manager region list.
- Click **Update**. *All of the regions should now be overlaid on the original image.*

12. Inspect Regions

- Use the ROI Manager to view each region in the original image by selecting each region label from the ROI Manager's list or by clicking the region in the image. If a region does not appear to be a complete, non-dividing nucleus, select it and click **Delete**, then continue to the next region.
- Once all the regions have been reviewed in this way, transfer the indices of the remaining good regions into the Python list below (section 3.3.3), setting the variable named **good_roi_exm_a_list**. The expected result for **expanded-hek-cells-a** is

```
good_roi_exm_a_list = range(2, 28) # regions 2 - 27 are good; region 1 is a partial nucleus
```

13. Annotate Original Image

- In the ROI Manager, click **Deselect**
- Click **Properties...**
 - **Range:** = *should automatically be filled in with the range of all the regions (e.g 1-26)*
 - **Position:** = none
 - **Group:** = none
 - **Stroke color:** = yellow
 - **Width:** = 4
 - **Fill color:** = none
 - **List coordinates** = NO
- In the main Fiji toolbar, select **Image > Overlay > Labels...**
 - **Color:** = yellow
 - **Font size** = 24 *adjust as needed*
 - **Show labels** = YES
 - **Use names as labels** = NO
 - **Draw backgrounds** = NO
 - **Bold** = NO
- In the ROI Manager, click **Deselect**
- In the ROI Manager, click **Flatten [F]** *this should open a new image window*
- In the main Fiji toolbar, select **File > Save As > PNG...** to save the new image
 - Save in: ~/ExM-Demo-Figures/data/nuc-size-analysis/

– File name: `expanded-hek-cells-a-ANNOTATED.png`

Repeat this process for each of the other nuclei images to be analyzed: `expanded-hek-cells-b.tif` and `normal-hek-cells.tif`.

3. Analysis, Plotting of Nuclei Sizes with Python

Run the following cells to process the extracted results data.

3.1 Package Imports

```
# 3.1.1 base and metaprogramming imports
import os
import sys
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib as mpl
import matplotlib.pyplot as plt

# 3.1.2 ipynb-specific imports
try:
    import ipynbname
except Exception as e:
    warnings.warn(f'`ipynbname` module import failed with error: {e!r}')
    ipynbname = None

if ipynbname is not None:
    _file = ipynbname.path()
else:
    # manually set the full path to the current notebook below if `ipynbname`
    # import fails
    _file = ''

# 3.1.3 key path variables and management
source_dir = os.path.split(_file)[0]
repo_dir = os.path.abspath(os.path.join(source_dir, '..'))
data_dir = os.path.join(repo_dir, 'data')
figure_dir = os.path.join(repo_dir, 'figures')
if source_dir not in sys.path:
    sys.path.append(source_dir)

print('Imports done.')
```

Imports done.

3.2 Plotting Defaults

```
# 3.2.1 load custom fonts, if available
custom_fonts = [
    'C:\\Windows\\Fonts\\Linotype - HelveticaLTPro-Black.otf',
    'C:\\Windows\\Fonts\\Linotype - HelveticaLTPro-Bold.otf',
    'C:\\Windows\\Fonts\\Linotype - HelveticaLTPro-Roman.otf',
    'C:\\Windows\\Fonts\\Linotype - HelveticaLTPro-Oblique.otf']
for font_path in custom_fonts:
    if os.path.exists(font_path):
        print(f'Adding font: "{os.path.split(font_path)[-1]}" to mpl.')
        mpl.font_manager.fontManager.addfont(font_path)

# 3.2.2 set plotting style
sns.set_style(
    'ticks',
    {'font.sans-serif': [
        'Helvetica LT Pro', 'Helvetica', 'Arial', 'sans-serif'],
     'mathtext.fontset': ['cm'],
     'axes.linewidth': 2})

# 3.2.3 set axis and figure defaults
axes_lw = 1.25
mpl.rcParams['axes.linewidth'] = axes_lw
mpl.rcParams['xtick.major.width'] = axes_lw
mpl.rcParams['xtick.minor.width'] = axes_lw
mpl.rcParams['ytick.major.width'] = axes_lw
mpl.rcParams['ytick.minor.width'] = axes_lw

mpl.rcParams['axes.facecolor'] = 'none'

mpl.rcParams['figure.frameon'] = False
mpl.rcParams['figure.facecolor'] = 'none'
mpl.rcParams['figure.edgecolor'] = 'none'

print('Plotting defaults set; done.')
```

Adding font: "Linotype - HelveticaLTPro-Black.otf" to mpl.

Adding font: "Linotype - HelveticaLTPro-Bold.otf" to mpl.

Adding font: "Linotype - HelveticaLTPro-Roman.otf" to mpl.

Adding font: "Linotype - HelveticaLTPro-Oblique.otf" to mpl.
Plotting defaults set; done.

3.3 Filepaths and Other Parameter Inputs

```
# 3.3.1 filepath names
# Edit these fields to match the files you wish to analyze.
nuc_size_dname = 'nuc-size-analysis'
normal_fname = 'normal-hek-cells-RESULTS.csv'
exm_a_fname = 'expanded-hek-cells-a-RESULTS.csv'
exm_b_fname = 'expanded-hek-cells-b-RESULTS.csv'

figure_name = 'nuc_size_comparison' # output figure name

# 3.3.2 table keys
blank_index_key = ' ' # Fiji populates the results `.csv` with a blank
                      # header for the first/index column
index_key = 'original_index'
category_key = 'category'
expanded_cat_name = 'expanded'
unexpanded_cat_name = 'normal'
file_key = 'original_file'
area_key = 'Area'
sqrt_area_key = '[Area]^(1/2)'

# 3.3.3 good region list (from manual inspection)
good_roi_normal_list = [
    9, 13, 15, 17, 19, 20, 21, 26, 28, 29,
    30, 31, 35, 36, 37, 38, 39, 40, 41, 43,
    44, 45, 46, 47, 48, 49, 51, 54, 55, 57,
    58, 59, 60, 61, 63, 65, 68, 69, 70, 75,
    76, 78, 79, 81, 82, 83, 84, 85, 86, 87,
    90, 92, 94, 96, 99, 103, 104, 105, 106, 109,
    111, 112, 113, 114, 115, 116, 117, 118, 119, 120,
    124, 128, 134, 137, 138, 142, 143, 144, 145, 148,
    150, 152, 153, 154, 155, 156, 163, 164, 166, 170,
    173, 174, 176, 177, 178, 180, 181, 184, 186, 188,
    189, 190, 191, 192, 193, 194, 198, 199, 200, 201,
    202, 204, 207, 209, 212, 213, 214, 215, 219, 220,
    221, 222, 224, 225, 228, 231, 234, 235, 236, 237,
    241, 242, 243, 246, 247, 249, 252, 254, 260, 264,
    265, 267, 268, 270, 280, 281, 287, 288]
```

```

good_roi_exm_a_list = range(2, 28) # 2 to 27, inclusive
good_roi_exm_b_list = range(1, 29) # all regions, 1 to 28, inclusive

print('Parameters set; done.')

```

Parameters set; done.

3.4 Loading and Processing Data Tables

```

# 3.4.1 generate full paths to files
nuc_size_dir = os.path.join(data_dir, nuc_size_dname)
normal_path = os.path.join(nuc_size_dir, normal_fname)
exm_a_path = os.path.join(nuc_size_dir, exm_a_fname)
exm_b_path = os.path.join(nuc_size_dir, exm_b_fname)

# 3.4.2 load `.csv` files into pandas `DataFrame`s
raw_normal_df = pd.read_csv(normal_path)
raw_exm_a_df = pd.read_csv(exm_a_path)
raw_exm_b_df = pd.read_csv(exm_b_path)

# 3.4.3 process the raw tables
def process_raw_df(raw_df, fname, cat_name, good_roi_list):
    df = raw_df.copy()
    df[file_key] = fname
    df[category_key] = cat_name
    df = df.rename(columns={blank_index_key: index_key})
    df = df[df[index_key].isin(list(good_roi_list))]
    return df.reset_index(drop=True)

normal_df = process_raw_df(
    raw_normal_df, normal_fname, unexpanded_cat_name, good_roi_normal_list)
exm_a_df = process_raw_df(
    raw_exm_a_df, exm_a_fname, expanded_cat_name, good_roi_exm_a_list)
exm_b_df = process_raw_df(
    raw_exm_b_df, exm_b_fname, expanded_cat_name, good_roi_exm_b_list)

# 3.4.4 make a concatenated table of all data points
nuc_size_df = (
    pd.concat([exm_a_df, exm_b_df, normal_df])
    .reset_index(drop=True))
nuc_size_df[category_key] = nuc_size_df[category_key].astype('category')

```



```
dtypes: float64(1), int64(1), object(2)
```

vvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvv Raw EXPANDED-A Data Frame: vvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvv

RangeIndex: 27 entries, 0 to 26

#	Column	Non-Null	Count	Dtype
---	--------	----------	-------	-------

```
dtypes: float64(1), int64(1)
```

0	1	1377.171
---	---	----------

```
vvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvv EXPANDED-A Data Frame: vvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvv
```

RangeIndex: 26 entries, 0 to 25

#	Column	Non-Null Count	Dtype
---	--------	----------------	-------

```
dtypes: float64(1), int64(1), object(2)
```

vvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvv Raw EXPANDED-B Data Frame: vvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvvv

RangeIndex: 28 entries, 0 to 27

#	Column	Non-Null	Count	Dtype
---	--------	----------	-------	-------

10

3.5 Show ROI Delete List

This is a tool to print the regions that were removed from the original ROI Manager at step 2.12 above.

```
# 3.5.1 helper function for printing a list as columns
def print_list_in_columns(data, num_columns=10, column_padding=2):
    # function partially generated by Gemini AI, prompted by Corban Swain
    max_len = max(len(str(item)) for item in data) if data else 0
    col_width = max_len + column_padding
    for i, item in enumerate(data):
        print(str(item).ljust(col_width), end="")
        if (i + 1) % num_columns == 0:
            print()
    if len(data) % num_columns != 0:
        print() # Print a final newline if the list doesn't perfectly fill the
                # last row

# 3.5.2 helper function for computing and printing the deleted regions
def print_delete_list(good_list, raw_df, name):
    good_len = len(good_list)
    total_len = len(raw_df)
    bad_len = total_len - good_len
    print(f'\nLabels of deleted ROIs for {name} dataset:')
    print(f'({good_len:d} regions kept; '
          f'{bad_len:d} regions deleted)')
    if bad_len == 0:
        print('none')
        return
    print_list_in_columns(
        [(i+1) for i in range(total_len) if (i+1) not in good_list])

# 3.5.2 printing the delete list for each sample
print_delete_list(good_roi_normal_list, raw_normal_df, 'NORMAL')
print_delete_list(good_roi_exm_a_list, raw_exm_a_df, 'EXPANDED-A')
print_delete_list(good_roi_exm_b_list, raw_exm_b_df, 'EXPANDED-B')

print('\nDelete lists printed; done.')
```

Labels of deleted ROIs for NORMAL dataset:

(148 regions kept; 149 regions deleted)

1	2	3	4	5	6	7	8	10	11
12	14	16	18	22	23	24	25	27	32

```

33  34  42  50  52  53  56  62  64  66
67  71  72  73  74  77  80  88  89  91
93  95  97  98  100 101 102 107 108 110
121 122 123 125 126 127 129 130 131 132
133 135 136 139 140 141 146 147 149 151
157 158 159 160 161 162 165 167 168 169
171 172 175 179 182 183 185 187 195 196
197 203 205 206 208 210 211 216 217 218
223 226 227 229 230 232 233 238 239 240
244 245 248 250 251 253 255 256 257 258
259 261 262 263 266 269 271 272 273 274
275 276 277 278 279 282 283 284 285 286
289 290 291 292 293 294 295 296 297

```

Labels of deleted ROIs for EXPANDED-A dataset:
(26 regions kept; 1 regions deleted)
1

Labels of deleted ROIs for EXPANDED-B dataset:
(28 regions kept; 0 regions deleted)
none

Delete lists printed; done.

3.6 Compute Median Nuclei Sizes and n

```

# 3.6.1 create a pivot table computing the median sqrt area for each category
pivot = nuc_size_df.pivot_table(
    columns=category_key,
    values=sqrt_area_key,
    aggfunc='median',
    observed=True)

# 3.6.2 create pivot table counting the number of nuclei for each category
exm_median = pivot[expanded_cat_name].item()
unexm_median = pivot[unexpanded_cat_name].item()
ex_factor = exm_median / unexm_median

count_pivot = nuc_size_df.pivot_table(
    columns=category_key,
    values=sqrt_area_key,
    aggfunc='count',

```

```

    observed=True)
count_pivot = count_pivot.rename(index={sqrt_area_key: 'n'})

exm_n = count_pivot[expanded_cat_name].item()
unexm_n = count_pivot[unexpanded_cat_name].item()

print('Median [Area]^(1/2):')
print(pivot)
print()
print('Num Data Points:')
print(count_pivot)
print()
print(f'Expansion Factor: {ex_factor:.1f}x')
print('\nData summaries generated; done.')

```

```

Median [Area]^(1/2):
category      expanded      normal
[Area]^(1/2)  34.59245  8.826096

```

```

Num Data Points:
category  expanded  normal
n          54      148

```

```

Expansion Factor: 3.9x

```

```

Data summaries generated; done.

```

3.7 Create and Save Figure

```

# 3.7.1 generate main figure object
fig = plt.figure(
    figsize=(2.5, 4.5),
    dpi=300)

# 3.7.2 set color palettes
palette_p = sns.color_palette(
    [sns.color_palette('pastel')[0],
     sns.color_palette('pastel')[3]])
palette_n = sns.color_palette(
    [sns.color_palette()[0],
     sns.color_palette()[3]])

```

3.7.3 use seaborn to plot boxplots with overlaid swarmplots from data

```
ax = sns.boxplot(  
    data=nuc_size_df,  
    x=category_key,  
    y=sqrt_area_key,  
    hue=category_key,  
    palette=palette_p,  
    order=[unexpanded_cat_name, expanded_cat_name],  
    hue_order=[unexpanded_cat_name, expanded_cat_name],  
    linecolor='k',  
    medianprops={'lw': 1.5})  
sns.swarmplot(  
    data=nuc_size_df,  
    x=category_key,  
    y=sqrt_area_key,  
    hue=category_key,  
    size=2.5,  
    alpha=0.3,  
    ax=ax,  
    palette=palette_n,  
    color='k',  
    order=[unexpanded_cat_name, expanded_cat_name],  
    hue_order=[unexpanded_cat_name, expanded_cat_name])
```

3.7.4 plot horizontal lines at the category medians

```
ax.axhline(  
    y=unexm_median,  
    color=palette_n[0],  
    ls='--',  
    lw=1,  
    alpha=0.75  
)  
ax.axhline(  
    y=exm_median,  
    color=palette_n[1],  
    ls='--',  
    lw=1,  
    alpha=0.75  
)
```

3.7.5 format look of axes

```

sns.despine(right=True, bottom=True, top=True)
ax.set_ylim(0, 50)
ax.spines['bottom'].set_bounds(0, 1)
ax.set_yticks(
    [0,
     unexm_median,
     25,
     exm_median,
     50])
ax.set_yticklabels(
    ['0',
     f'{unexm_median:.1f}',
     '25',
     f'{exm_median:.1f}',
     '50'])
ax.set_ylabel('\u221A nucleus area (\u00b5m)')
ax.set_xlabel(None)
ax.tick_params(axis='x', length=0, pad=-4)

# 3.7.6 annotate expansion factor
ax.annotate(
    f'{ex_factor:.1f}\u00D7',
    xy=(0.5, exm_median),
    xytext=(0.5, 20.0),
    arrowprops=dict(
        arrowstyle='->',
        color='black',
    ),
    annotation_clip=False,
    ha='center',
    va='center',
    size=14,
    weight='bold',
)
ax.annotate(
    f'{ex_factor:.1f}\u00D7',
    xy=(0.5, unexm_median),
    xytext=(0.5, 20.0),
    arrowprops=dict(
        arrowstyle='-',
        color='black',

```



```

    ),
    annotation_clip=False,
    ha='center',
    va='center',
    alpha=0,
    size=14,
    weight='bold',
)

# 3.7.8 annotate num data points
ax.text(
    0, 2,
    f'n = {unexm_n}',
    ha='center',
    va='baseline',
    size=7.5,
    style='oblique',
    alpha=0.75)
ax.text(
    1, 2,
    f'n = {exm_n}',
    ha='center',
    va='baseline',
    size=7.5,
    style='oblique',
    alpha=0.75)

# 3.7.9 helper function for saving figure to pdf
def save_figures(
    filename=None,
    figs=None,
    dpi=200,
    fmt='pdf',
    directory=None,
    add_filename_timestamp=True,
    stamp_kwargs=None):
    from matplotlib.backends.backend_pdf import PdfPages

    if filename is None:
        filename = 'all_figures'

```

```

if add_filename_timestamp:
    filename = filename + '_' + time.strftime('%y%m%d-%H%M')

if figs is None:
    figs = [plt.figure(n) for n in plt.get_fignums()]
else:
    try:
        _ = iter(figs)
    except TypeError:
        figs = [figs, ]

if stamp_kwargs:
    [stamp_fig(fig, **stamp_kwargs) for fig in figs]

directory = 'figures' if directory is None else directory
try:
    os.mkdir(directory)
except FileExistsError:
    pass

file_path = os.path.join(directory, filename)
if fmt == 'pdf':
    file_path += '.pdf'
    with PdfPages(file_path) as pp:
        for fig in figs:
            fig.savefig(pp, format='pdf', dpi=dpi)
else:
    for i, fig in enumerate(figs):
        fig.savefig('%s_%d.%s' % (file_path, i, fmt), format=fmt, dpi=dpi)

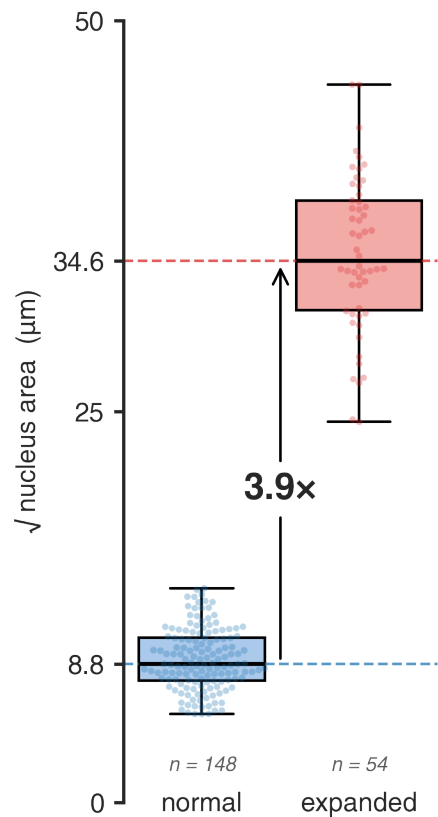
# 3.7.10 perform final layout of plot
plt.tight_layout()

# 3.7.11 export figure to pdf
save_figures(
    filename=figure_name,
    add_filename_timestamp=False,
    directory=figure_dir)

print('Figure generated and saved; done.')

```

Figure generated and saved; done.



4 Notebook Conversion

This notebook can be converted to a **.pdf** via the following **pandoc** (<https://pandoc.org/>) command:

for *Unix-style command line*:

```
pandoc -o "supplementary_file_03-nuc-area.pdf" \
  --pdf-engine=xelatex --highlight-style tango \
  -V geometry:landscape \
  -V geometry:top=0.75in,bottom=0.75in,left=1in,right=2in
  supplementary_file_03-nuc-area.ipynb
```

for *Windows Command Prompt*:

```
pandoc -o "supplementary_file_03-nuc-area.pdf" ^
  --pdf-engine=xelatex --highlight-style tango ^
```

```
-V geometry:landscape ^  
-V geometry:top=0.75in,bottom=0.75in,left=1in,right=2in ^  
supplementary_file_03-nuc-area.ipynb
```